founders of quantum physics. But Zeilinger, through his experimental work, has been determined to show that there is no limit to the validity of quantum physics.

But this leaves a question: If only classical things exist objectively, yet quantum physics applies to everything, what is classical? More generally, how can we account for the world we see around us? According to Zeilinger, our everyday world is classical—but quantum physics must also accurately describe what we see in our everyday lives, because there's no boundary to its validity. How can we form a coherent picture of reality from this version of the Copenhagen interpretation? Zeilinger's answer to this question is unexpectedly simple. "I don't know what you mean by that," he said. "I don't think that you can even define it precisely."

What the hell is going on here?

Not all physicists agree with Zeilinger. "The Copenhagen interpretation assumes a mysterious division between the microscopic world governed by quantum mechanics and a macroscopic world of [measurement] apparatus and observers that obeys classical physics," said Steven Weinberg, winner of the 1979 Nobel Prize in Physics. "This is clearly unsatisfactory. If quantum mechanics applies to everything, then it must apply to a physicist's measurement apparatus, and to physicists themselves. On the other hand, if quantum mechanics does not apply to everything, then we need to know where to draw the boundary of its area of validity. Does it apply only to systems that are not too large? Does it apply if a measurement is made by some automatic apparatus, and no human reads the result?" Gerard 't Hooft, winner of the 1999 Nobel Prize in Physics, struck a more conciliatory note. "I go along with everything [Copenhagen] says, except for one thing, and the one thing is you're not allowed to ask any questions," he says. "Or to be more precise, there are certain questions you should not ask. I say, no, I'm going to ask them anyway. You don't want me to ask questions? Sorry. I have the strong impression that there is much more to say, and that asking questions is useful." And Sir Anthony Leggett, winner of the 2003 Nobel Prize in Physics, said that he must make "an awful confession: If you were